A Motion-Capture Driven Humanoid Hand for Dance and Cultural Preservation

Guneet Kapur

Abstract

Dance is a universal language of expression used across cultures to pass down stories, values, and traditions. As generations pass, culturally significant dance gestures risk being changed or lost due to the absence of a reliable, embodied preservation method. To address this problem, I created a 3D-printed humanoid robotic hand controlled by an Arduino Nano microcontroller. This robotic hand takes live hand movements captured from a motion-capture system, converts those movements into joint-angle data, and then physically reproduces those same positions. The motion data is then stored in a CSV file as mapped joint angles, allowing the robotic hand to reproduce the recorded gesture consistently. A website database was also created to enable users to contribute and archive cultural hand gestures, supporting long-term preservation. In order to ensure the robotic hand is accurate in its replication, specific hand gestures for each dance form and each gesture underwent 10 sets of trials. To quantify the accuracy of replication, a testing protocol was designed to calculate angular deviation ($|\theta_{robot} - \theta_{cam}|$) and a Pose Similarity score (0-1). Across all trials, the robotic hand achieved a, very low, total average angular deviation of 1.72° and an overall Pose Similarity score of 0.990. The project also open-sources the robotic hand designs and the software to generate the motion capture data, which ultimately serves to preserve an accurate repository of global cultural hand movements for future generations.

Introduction

Dance is a form of expression that many people enjoy in their daily lives, whether it's their livelihood, their favorite pastime, or simply a form of entertainment. The language of dance is universal, with the ability to convey emotion and meaning without words. In fact, in many cultures, dance plays a key role in keeping traditions alive, as well as passing down stories and values to the next generation. In the Indian classical dance form called Kathak, an essential part of the performance is the intricate hand gestures called *mudras*. These mudras are very specific finger movements that aid in preserving the stories passed down (Saxena, 1982). However, it is not only dance forms from India that use hand gestures to pass down stories; it is cultures from all around the world. Seen in Polynesian traditions such as the Hawaiian Hula, Ori Tahiti, and Sasa; Indian classical forms of Bharatnatayam and Kathak; and Indigenous cultures including the Māori Haka of New Zealand, the Baishou Dance of China, and Plains Indian dances of North America, hand movements in dance are essential to storytelling (Tulio, 2022).



Figure 1: Example of a specific cultural hand movement (mudra) from India
Published data by youtube.com

The existing research shows the need for an improved method of preserving and passing down cultural dances undisturbed. As of now, the only reliable way to pass down accurate cultural dance movements is from teacher to student, as dance relies heavily on person-to-person transmission. Although YouTube and other platforms are now widely accessible for aspiring learners, videos posted are not necessarily reliable or accurate enough for sufficient structured learning, especially when it comes to intricate forms of dance (Kang et al. 2023). In short, online methods lack the immersive and embodied experience of in-person training (Li et al. 2025). Additionally, the literature highlights how dance transforms from a "first existence" to a "second

existence” when taught, meaning that as dance is passed down from generation to generation, it is inevitably reshaped over time (Georgios et al. 2017). Unlike written language, there is currently no permanent and reliable way to pass down dance movements unchanged.

This gap inspired the beginning of my project. I created a way to permanently preserve original cultural dance movements for future generations. My project is a humanoid robotic hand that can store and physically reproduce the expressive hand gestures used in classical and cultural dances. My research question is: How accurately can a humanoid robotic hand reproduce recorded dance hand gestures using motion-capture data?

The primary goal of this project was to create a program that is able to capture choreographed hand gestures from a live video stream and physically replicate it. This set of saved gestures will become a large database of gestures, forming a “digital archive.” It will store data so that the hand gestures can be preserved, replayed, studied, and taught with the original movements unchanged.

My secondary goal was to develop a project site on which anyone can access the 3D motion-capture, playback, and movement-execution code that powers the system, along with the files needed to 3D print the robotic hand. People will record their own hand gestures to add to the database. This platform will be used as an open-source, cross-cultural repository for hand dance movements so people can contribute to and learn from all the different gestures saved on the archive.

Methods and Materials

To start, I 3D-printed the physical model of the humanoid robotic hand utilizing open-source STL files online (Gross, 2017). I then assembled my hand using a video posted online by the maker of the open-source files. To create the full function of the hand, I used fishing line (for the "tendons"), screws, servos and servo attachments, wiring, breadboards, an Arduino Nano microcontroller, and an Arduino 5V power source.

The second part of my project is the motion-capture component. Utilizing MediaPipe software from Google and OpenCV's open-source libraries, I was able to create a program run straight from my Macbook which would open the camera to capture the hand angles. Shown to the right in Figure 3, this motion-capture system tracks key joint-angles of the hand that correspond to bending points of the fingers and thumb. Instead of storing angles relative to the hand's position within the camera frame, I coded the program to compute the angles between joint-angles. Therefore, the data saved represents finger bend angles relative to other points on the hand rather than based on their position in the screen, greatly improving accuracy. In the top left corner of the camera, the names of the fingers are displayed and the dynamic joint angles that will be saved are shown, ready to be translated over to the Arduino controlling the hand. Within the motion-capture camera software, I encoded multiple commands to make saving different hand gestures as efficient as possible: 1) A "label" (L) button which opens a text box in which you would type the name of the label you would like to save, 2) A "save" (S) button which allows you to save the hand angles that are seen on the screen, and 3) A "quit" (Q) button which allows you to exit the camera program at any point. The gestures are then saved to a CSV

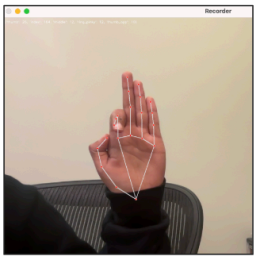


Figure 3: Motion-capture interface used to record hand gestures Produced by the student.

file where they are stored and then displayed in a numbered list when playing back a gesture on the robotic hand.

The third part of my project was to create the website mentioned in the Introduction. Using the help of Cursor AI, I built a website which makes it easy and convenient to play gestures on the hand. As shown in Figure 4, to the left of the screen, the list of current gestures saved in the CSV file are displayed and numbered 1, 2, 3, etc. At the bottom of the screen, there is a “record a gesture” button, which runs the software that opens the motion capture camera. To the right of the screen, there is a text box prompt section in which a person can enter number labels from the list of gestures displayed on the left. There are two buttons at the bottom of this box. First, “Save as a Group”, which will take the numbers of the gestures inputted in the text box and save them as another number in the list of gestures. When the number of the group is chosen to be played on the robotic hand, the gestures saved within the group will play consecutively. Second, the "Play" button plays the gesture(s) written in the text box. In order to advance the website and fulfill its goal of being a fully open-source repository in which many people can contribute and add their own hand gestures to the CSV file, the gestures will be sorted into “categories”. Each category will represent a different culture, and the hand gestures saved can be sorted into each category based on the person inputting the gestures. In order to validate the reliability and consistency of the robotic hand,

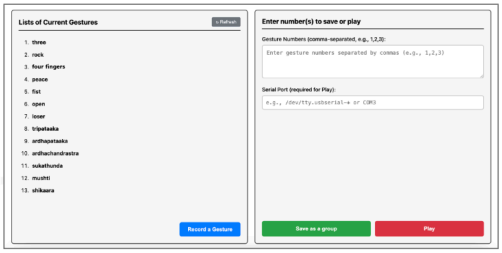


Figure 4: User website for selecting and saving recorded hand gestures
Produced by the student.

Results

A total of 25 trials were conducted across five distinct hand gestures ranging in complexity (G1-G5), with each gesture recorded once and executed five, consecutive, separate times. For each trial and finger, an angular deviation was calculated using $|\theta_{robot} - \theta_{cam}|$ and averaged to obtain a trial-level deviation. Using the trial-level angular deviations, a Total Average Angular Deviation (T) was calculated. Pose similarity was then calculated by doing one minus the percent error ($S = 1-(T/180)$). An example of the data collected (each finger's angular deviation) is shown in Figure 5. The T value was calculated in order to determine whether the robotic hand would be able to consistently reproduce the same hand gesture and the Pose Similarity Score allows for an overall estimate of the hand's reliability.

	Name	Thumb	Index	Middle	Ring_pinky	Thumb_opp
Original Gesture Degrees (Goal for each finger)		2	177	177	1	180
G1: 1 servo	Three	|0-2|	|180-177|	|180-177|	|0-1|	|180-180|
		|0-2|	|180-177|	|180-177|	|1-1|	|180-180|
		|0-2|	|180-177|	|180-177|	|2-1|	|180-180|
		|0-2|	|180-177|	|180-177|	|0-1|	|180-180|
		|0-2|	|180-177|	|180-177|	|0-1|	|180-180|
Original Gesture Degrees (Goal for each finger)		150	179	178	177	90
G2: 2 servos	Four	|147-150|	|180-179|	|180-178|	|180-177|	|89-90|
		|148-150|	|180-179|	|180-178|	|180-177|	|89-90|
		|148-150|	|180-179|	|180-178|	|180-177|	|89-90|
		|151-150|	|180-179|	|180-178|	|180-177|	|92-90|
		|150-150|	|180-179|	|180-178|	|180-177|	|92-90|

Figure 5: Example of data collected and used to calculate angular deviation.
Produced by the student.

My hypotheses were that the robotic hand would be able to accurately and consistently reproduce recorded hand gestures across repeated trials, supporting its use as a reliable "living archive" for cultural dance movements. I expected that simpler hand gestures with fewer simultaneous servos moving would exhibit lower angular deviation and greater consistency across trials, while more complex hand gestures involving multiple servos moving may show higher deviation.

Across all trials, the robotic hand achieved a T value of 1.72° and an overall Pose similarity score of 0.990. Mean angular deviation increased as gesture complexity increased. Gestures involving fewer active servos produced lower deviation values, while gestures involving more simultaneous finger movements produced higher deviation values. Observed servo angles remained closely aligned with expected values across all gestures and trials.

Standard deviation remained low across trials, indicating consistent reproduction of recorded poses. Simpler gestures showed the smallest spread in deviation, while more complex gestures showed slightly greater variability.

Discussion

The results prove that the robotic hand is both reliable and consistent over multiple trials for reproducing recorded hand gestures. The extremely low angular deviation is extremely important because in order for the robotic hand to be utilized as a learning tool, the repeated moves need to be virtually indistinguishable. The Pose Similarity score of 0.990 also demonstrates close agreement between the hand positions captured by the motion-capture system and the positions executed by the robotic hand.

There was also a clear pattern between gesture complexity and reproduction error. As the number of simultaneously activated servos increased, mean angular deviation increased. This supports the hypothesis that more complex gestures would produce greater error than simpler gestures. One likely explanation is that as additional fingers move at the same time, the system becomes more sensitive to cumulative mechanical error. Even with this increase in deviation, the deviation remained small across all gestures. This is important because the purpose of the system is not simply to move, but to preserve culturally meaningful gesture forms with enough consistency that they are visually indistinguishable from the original movements for learning and replay purposes.

There were two main limitations with this project. First, only 25 trials were conducted for the data analysis which limits the statistical depth of the results. Second, the system was tested only on one person’s hand gestures, which potentially changes the results when it comes to more people contributing.

Overall, the project suggests that humanoid robotics can contribute to cultural preservation by creating a stable, embodied archive of movement rather than relying only on observation-based transmission.

Conclusion

In conclusion, this project demonstrates that a motion-capture driven robotic hand can accurately and consistently reproduce cultural hand gestures. The mechanical system achieved low angular deviation and high pose similarity across all trials, confirming its reliability as a repeatable gesture reproduction mechanism. These results support the use of this robotic system as an embodied, “living” archive for preserving cultural hand gestures and supports the use of the system over a longer period of time. In the future, an improvement to the robotic hand would be to add a third axis of movement, such as a wrist movement. That would help to enhance the hand and capture even more of the vibrancy of cultural dance movements.

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